1. A discrete-time system has impulse response

$$h[n] = \delta[n] + (1/2)^n u[n] + (-5)^n u[n] \,.$$

What is the region of convergence of $H(z)$?

(A) $|z| > 5$

(B) $Re(z) > 5$

(C) $|z| < 1$

	Signal, $x(n)$	z-Transform, $X(z)$	ROC
1	$\delta(n)$	1	All z
2	$u(n)$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
3	$a^n u(n)$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
4	$na^n u(n)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
5	$-a^n u(-n-1)$	$\frac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
6	$-na^n u(-n-1)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$
7	$(\cos \omega_0 n)u(n)$	$\frac{1-z^{-1}\cos\omega_0}{1-2z^{-1}\cos\omega_0+z^{-2}}$	$\|z\| > 1$
8	$(\sin \omega_0 n)u(n)$	$\frac{z^{-1}\sin\omega_0}{1-2z^{-1}\cos\omega_0+z^{-2}}$	$\|z\| > 1$
9	$(a^n \cos \omega_0 n)u(n)$	$\frac{1-az^{-1}\cos\omega_0}{1-2az^{-1}\cos\omega_0+a^2z^{-2}}$	$\|z\| > \|a\|$
10	$(a^n \sin \omega_0 n)u(n)$	$\frac{az^{-1}\sin\omega_0}{1-2az^{-1}\cos\omega_0+a^2z^{-2}}$	$\|z\| > \|a\|$

$$H(z) = 1 + \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1+5z^{-1}}$$

2. A system has transfer function

$$H(z) = \frac{(1 - \frac{1}{4}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 + 3z^{-1})}$$

Is the system stable? $z = \frac{1}{2}$ $z = -3$

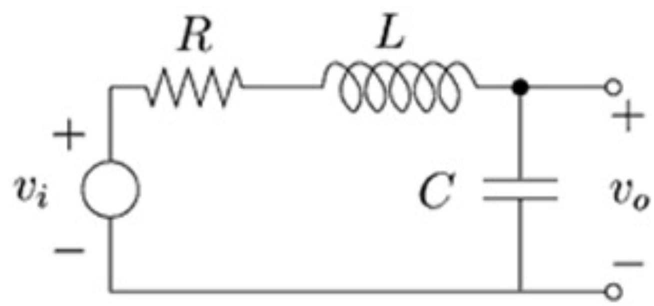
(A) Yes

(B) No

(C) It is stable for $|z| > 3$.

3. A circuit has transfer function

$$H(s) = \frac{1}{s^2LC + sRC + 1}.$$



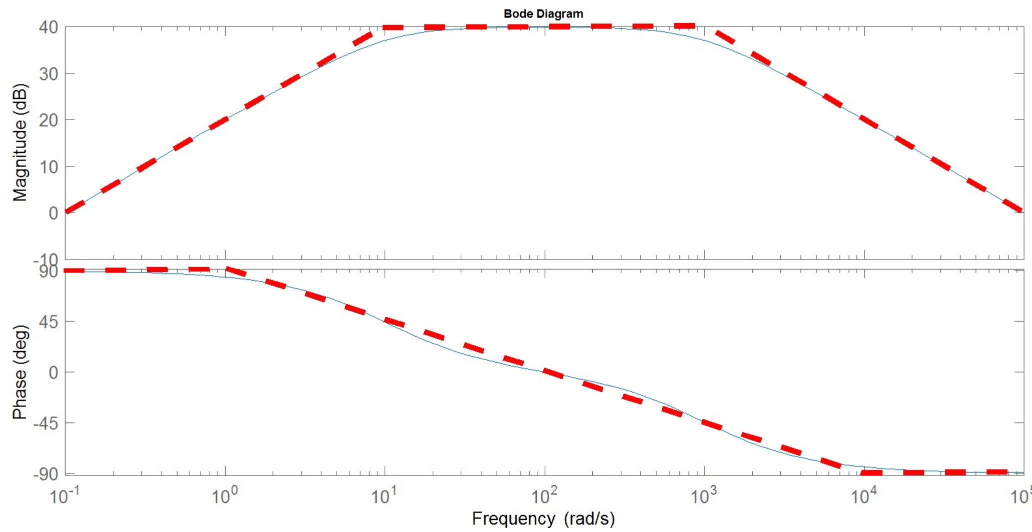
What is the (possible) resonant frequency of the circuit?

(A) $\omega_0 = \sqrt{LC}$

(B) $\omega_0 = \frac{1}{\sqrt{LC}}$

(C) $\omega_0 = R\sqrt{\frac{C}{L}}$

4. What is the transfer function of the system with Bode diagram



(A) $H(s) = \dfrac{10^2(s+1)}{(s+10)(s+1000)}$

(B) $H(s) = \dfrac{10^3}{s(s+10)(s+1000)}$

(C) $H(s) = \dfrac{10^5 s}{(s+10)(s+1000)}$

1. A discrete-time system has impulse response

$$h[n] = \delta[n] + (1/2)^n u[n] + (-5)^n u[n]\,.$$

What is the region of convergence of $H(z)$?

(A) $|z| > 5$

(B) $Re(z) > 5$

(C) $|z| < 1$

Solution:

$$H(z) = \mathcal{Z}\{h[n]\} = \mathcal{Z}\{\delta[n]\} + \mathcal{Z}\left\{\left(\tfrac{1}{2}\right)^n u[n]\right\} + \mathcal{Z}\{(-5)^n u[n]\}$$

$$= \underbrace{1}_{\text{(for all } z)} + \underbrace{\frac{1}{1 - \frac{1}{2} z^{-1}}}_{\left(\text{for } |z| > \frac{1}{2}\right)} + \underbrace{\frac{1}{1 + 5 z^{-1}}}_{(\text{for } |z| > 5)}$$

most "restrictive"

$\Rightarrow$ ROC of $H(z)$: $|z| > 5$

2. A system has transfer function

$$H(z) = \frac{(1 - \frac{1}{4}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 + 3z^{-1})}$$

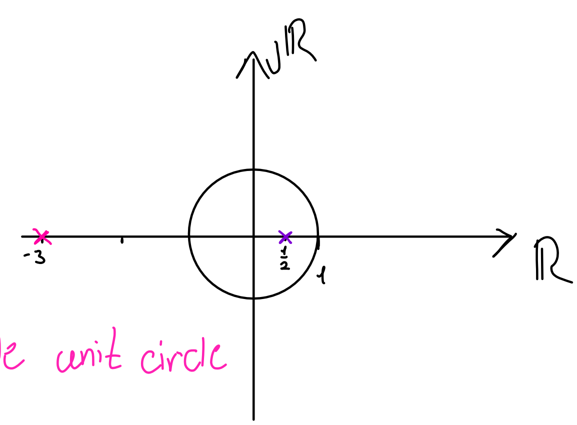
Is the system stable?

(A) Yes

(B) No

(C) It is stable for $|z| > 3$.

Solution: poles of $H(z)$: $\underline{z = \frac{1}{2}}$ and $\underline{z = -3}$

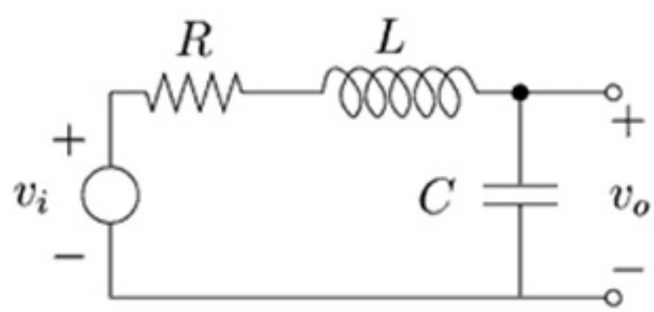


$\Rightarrow H(z)$ is unstable

pole outside unit circle

3. A circuit has transfer function

$$H(s) = \frac{1}{s^2LC + sRC + 1}.$$



What is the (possible) resonant frequency of the circuit?

(A) $\omega_0 = \sqrt{LC}$

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(C) $\omega_0 = R\sqrt{\frac{C}{L}}$

Solution: (also already given slides for Lectures 12-13)

compare

$$s^2 LC + s RC + 1 \longleftrightarrow s^2 T^2 + s(2\xi T) + 1$$

$$LC = T^2 \rightarrow T = \sqrt{LC}$$

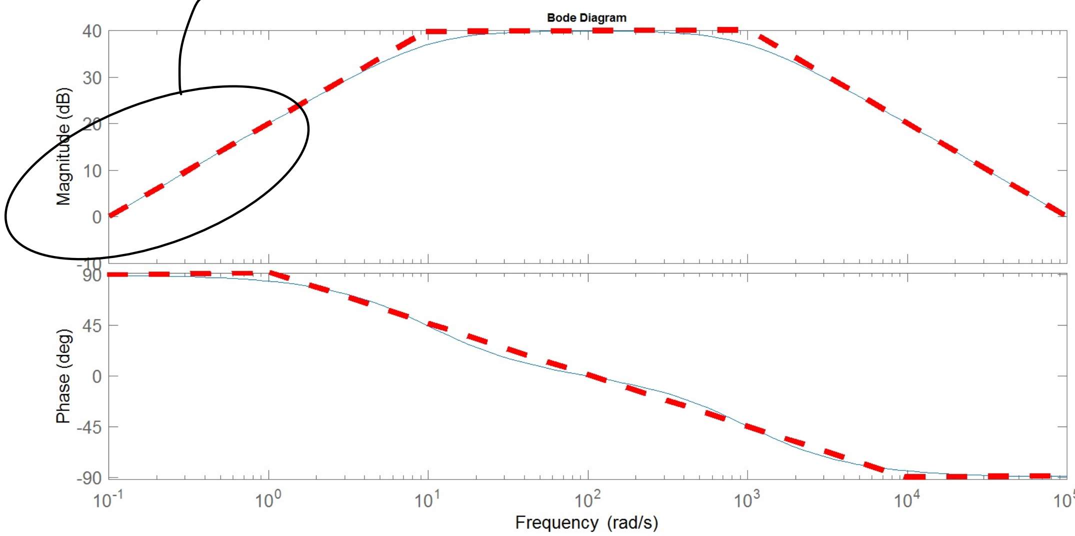
$$RC = 2\xi T \rightarrow \xi = \frac{RC}{2T} = \frac{1}{2} R \sqrt{\frac{C}{L}}$$

The resonance frequency (also called natural frequency) is

$$\omega_0 = \frac{1}{T} = \frac{1}{\sqrt{LC}}$$

4. What is the transfer function of the system with Bode diagram

initial slope indicates zero at origin → C is only feasible answer

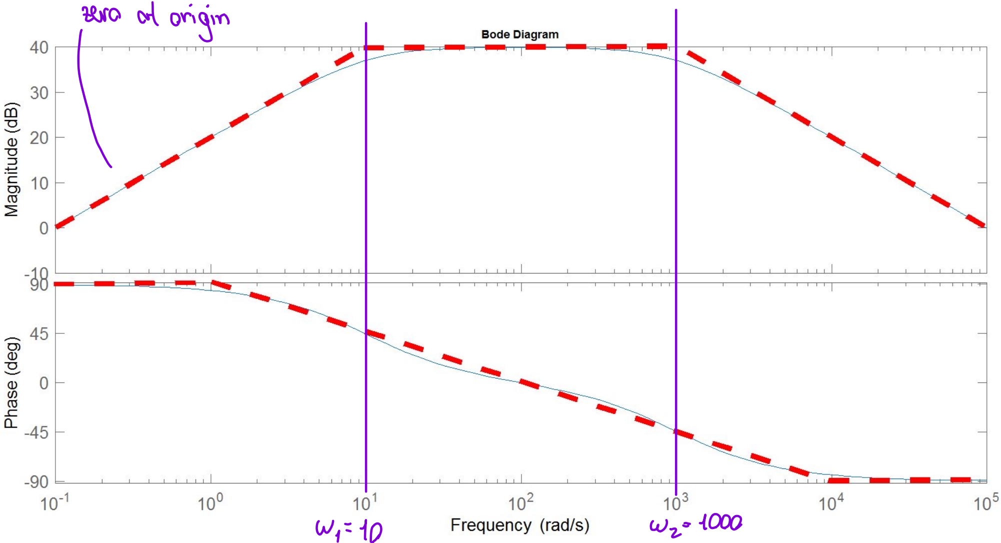


(A) $H(s) = \dfrac{10^2(s+1)}{(s+10)(s+1000)}$

(B) $H(s) = \dfrac{10^3}{s(s+10)(s+1000)}$

(C) $H(s) = \dfrac{10^5 s}{(s+10)(s+1000)}$

However, it is a good exercise to determine H(s) from the diagram:



zero at origin
Bode Diagram
Magnitude (dB)
40
30
20
10
0
-10
Phase (deg)
90
45
0
-45
-90
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
Frequency (rad/s)
$\omega_1 = 10$
$\omega_2 = 1000$

at / around $\omega_1 = 10$: slope of $|H(j\omega)|$ changes by -20 dB/dec

$\angle H(j\omega)$ changes by -90° between $\omega = 10^0$ and $\omega = 10^2$

↳ LHP pole: $s = -10$

at / around $\omega_2 = 10^3$ slope of $|H(j\omega)|$ changes by -20 dB/dec

$\angle H(j\omega)$ changes by -90° between $\omega = 10^2$ and $\omega = 10^4$

↳ LHP pole: $s = -1000$

$$H(s) = K \frac{s}{\left(1 + \frac{1}{10}s\right)\left(1 + \frac{1}{1000}s\right)}$$

$$= 10 \frac{s}{\left(1 + \frac{1}{10}s\right)\left(1 + \frac{1}{1000}s\right)}$$

$$\boxed{H(s) = \frac{10^5 s}{(s+10)(s+1000)}}$$

find K: $\left. |H(j\omega)| \right|_{\omega = 0.1} = 0\,\text{dB} = 1$

$$1 = \left| K \frac{0.1j}{\underbrace{\left(1 + \frac{1}{10}0.1j\right)\left(1 + \frac{1}{1000}0.1j\right)}_{1 + 0.01j + 0.0001j - 10^{-6}}} \right| \approx |K \cdot 0.1j|$$

$\Rightarrow K = 10$